MATH 215 EXAM 2 L10–L18 (sheet tags explained here)

5R6 = exam equation sheet **page 5, Right column, 6th formula** from the top (L = left column). **5C1**–**5C6** = p5 centered coordinate-conversion rows; **5top** = p5 cross-product determinant; **6B** = p6 projectile bullets; **6M1**–**6M5** = p6 full-width rows below the bullets ($\int \mathbf{r}$, $\int_a^b \mathbf{r}$, s 2D, s 3D, $s(t)$); **p3** = derivative | integral table; **p1** trig identities (by name); **p2** unit circle; **p4** = Unit 1 (vectors, lines, planes). ★= shortcut/extra, ⚠= pitfall, *Ex* = worked example with answer.

L10 QUADRIC SURFACES (not on the sheet — all here)

▸**Identify a surface:** look at the powers. **1) A variable is missing ⇒ cylinder**: draw the curve in the plane of the 2 variables present, slide it parallel to the missing variable's axis (lines = rulings). $x^2+z^2=25$: circle ⇒ tube along the y-axis; $y=\sin x$: wavy sheet, rulings $\parallel z$; $z=x^2$: parabolic trough along y. **2) One variable only to the 1st power ⇒ paraboloid**, axis = that linear variable: squares with the **same sign ⇒ elliptic paraboloid** (bowl) $z=x^2/a^2+y^2/b^2$; **opposite signs ⇒ hyperbolic paraboloid** (saddle) $z=x^2/a^2-y^2/b^2$. **3) All three squared**: constants right, divide so RHS = 1: $\pm x^2/a^2 \pm y^2/b^2 \pm z^2/c^2=1$, then count minus signs: **0 ⇒ ellipsoid** (egg; intercepts $\pm a,\pm b,\pm c$; $a=b=c$ ⇒ sphere); **1 ⇒ hyperboloid of one sheet** (hourglass; axis = the minus variable); **2 ⇒ hyperboloid of two sheets** (two bowls; axis = the plus variable; no points for $|z|<c$ if axis z); 3 ⇒ no points. **RHS** = 0 with mixed signs ⇒ **elliptic cone** (vertex at center, axis = the odd-sign variable). RHS = −1? multiply by −1 first. $a,b,c>0$ are semi-axes = $\sqrt{\text{denominators}}$ ($x^2/4 \Rightarrow a=2$). ★Axis = the odd one out (the linear variable or the lone sign).

▸**Traces** (surface ∩ a plane: fix one variable). Ellipsoid: ellipses every way. 1 sheet: ⊥ axis ellipses, ∥ axis hyperbolas. 2 sheets: ⊥ axis ellipse or nothing, ∥ hyperbolas. Cone: ⊥ axis ellipses (point at vertex), ∥ hyperbolas; coordinate planes containing the axis ⇒ 2 crossing lines. Elliptic parab.: ⊥ ellipses, ∥ parabolas. Hyperbolic parab.: ⊥ hyperbolas (2 lines at vertex level), ∥ parabolas opening opposite ways. **How:** sub the plane (xz-plane ⇒ $y=0$; $yz \Rightarrow x=0$; $xy \Rightarrow z=0$; "plane $x=3$" ⇒ $x=3$), simplify, name the 2D curve: $u^2/p^2+v^2/q^2=1$ ellipse (circle if $p=q$); $u^2/p^2-v^2/q^2=1$ hyperbola opening along u (the + one); $v=ku^2$ parabola; squares = 0 ⇒ point (sum) or 2 lines (difference); sum of squares = negative ⇒ empty. *Ex* $z^2/25+x^2/9-y^2/81=0$ in $x=3$: $z^2/25+1-y^2/81=0 \Rightarrow y^2/81-z^2/25=1$ (hyperbola along y). *Ex* $x^2/4+y^2/25+z^2/49=1$ in xz-plane: ellipse $x^2/4+z^2/49=1$.

▸**Standard form:** divide by the constant: $3x^2+12y^2+4z^2=12$ (÷12) ⇒ $x^2/4+y^2/1+z^2/3=1$. Coefficient k on x^2 ⇒ denominator $1/k$ ($8x^2=x^2/(1/8)$).

▸**Center / vertex (complete the square):** group by variable; **factor out each x^2 coefficient**; $x^2+bx=(x+b/2)^2-b^2/4$; multiply what you subtracted by the factored coefficient; constants to the right. $(x-h)^2 \Rightarrow$ center h (⚠ sign flips: $(x+1)^2 \Rightarrow h=-1$). Paraboloid: isolate the linear variable and factor its coefficient ($24z+24=24(z+1) \Rightarrow l=-1$); vertex = "center". *Ex* (Quiz 3) $9x^2+4y^2+36z^2+18x-8y-72z+13=0$: $9(x+1)^2-9+4(y-1)^2-4+36(z-1)^2-36+13=0 \Rightarrow 9(x+1)^2+4(y-1)^2+36(z-1)^2=36$: ellipsoid, center $(-1,1,1)$. *Ex* $4x^2-6y^2-8x+24y-24z-20=0$: $4(x-1)^2-4-6(y-2)^2+24-24z-20=0 \Rightarrow z=(x-1)^2/6-(y-2)^2/4$: saddle, center $(1,2,0)$. *Ex* $3x^2+5y^2+6x+20y-15z+23=0 \Rightarrow z=(x+1)^2/5+(y+2)^2/3$: center $(-1,-2,0)$.

L11 CYLINDRICAL & SPHERICAL COORDINATES

Cylindrical (r,θ,z): r = distance from the z-axis (≥ 0; polar radius of the shadow (x,y)); θ = angle in the xy-plane, CCW from $+x$; z = ordinary height. **Spherical** (ρ,θ,ϕ): ρ = distance from the origin (≥0); θ = same as cylindrical; ϕ = angle **down from the $+z$-axis**, $0\le\phi\le\pi$ ($\phi<\pi/2$ above the xy-plane, $=\pi/2$ on it, $>\pi/2$ below). Links: $r=\rho\sin\phi$, $z=\rho\cos\phi$, $\rho^2=r^2+z^2=x^2+y^2+z^2$.

▸**Convert a point:** pick the row: rect⇒cyl **5C1**; cyl⇒rect **5C2**; rect⇒sph **5C3**; sph⇒rect **5C4**; sph⇒cyl **5C5**; cyl⇒sph **5C6**. ⚠ **θ quadrant fix** ($\tan^{-1}$ only returns QI/QIV): sketch the signs of (x,y) first. $x>0, y\ge 0$: $\theta=\tan^{-1}(y/x)$; $x<0$ (QII, QIII): $\tan^{-1}(y/x)+180°$ $(+\pi)$; $x>0, y<0$ (QIV): $+360°$ $(+2\pi)$; $x=0$: 90° if $y>0$, 270° if $y<0$. $\phi=\cos^{-1}(z/\rho)$ needs **no** fix ($\cos^{-1}\in[0,\pi]$). ⚠ **"$\theta=\theta$": copy θ exactly as given** (Quiz 3 lost a point turning $-\pi/2$ into $3\pi/2$). Rationalize: $-1/\sqrt{82}=-\sqrt{82}/82$. "Exact" = radicals and π; "degrees, 2 dp" = decimals (calculator in degree mode). *Ex* $(-4,-8,6)$⇒sph: $\rho=\sqrt{116}\approx 10.77$, $\theta=\tan^{-1}2+180°=243.43°$, $\phi=\cos^{-1}(6/\sqrt{116})=56.15°$. *Ex* $(-1,4,-6)$⇒cyl: $(\sqrt{17}, 104.04°, -6)$ [QII: $\tan^{-1}(-4)+180°$]. *Ex* cyl $(9,-\pi/2,-1)$⇒sph: $(\sqrt{82}, -\pi/2, \cos^{-1}(-\sqrt{82}/82))$. *Ex* cyl $(4,3\pi/4,-1)$⇒rect: $(-2\sqrt{2}, 2\sqrt{2}, -1)$. *Ex* sph $(10,\pi/3,5\pi/4)$⇒rect: $(-5\sqrt{2}/2, -5\sqrt{6}/2, -5\sqrt{2})$.

▸**Convert an equation: cyl⇒rect**: make $r\cos\theta$ $(=x)$, $r\sin\theta$ $(=y)$, r^2 $(=x^2+y^2)$ appear — **multiply both sides by** r if needed; $\sec\theta=1/(\cos\theta)$. *Ex* $r=10\sec\theta \Rightarrow r\cos\theta=10 \Rightarrow x=10$ (a plane). *Ex* $r=8\sin\theta \Rightarrow r^2=8r\sin\theta \Rightarrow x^2+y^2=8y$. *Ex* $r^2+(z-4)^2=50 \Rightarrow x^2+y^2+(z-4)^2=50$. ★**Double angle p1**: $r^2\cos 2\theta=r^2(\cos^2\theta-\sin^2\theta)=x^2-y^2$; $r^2\sin 2\theta=2(r\cos\theta)(r\sin\theta)=2xy$. **rect⇒cyl**: $x^2+y^2\to r^2$, $x\to r\cos\theta$, $y\to r\sin\theta$, then solve for r (dividing by r is fine). *Ex* $x^2+(y-4)^2=16 \Rightarrow x^2+y^2-8y=0 \Rightarrow r^2=8r\sin\theta \Rightarrow r=8\sin\theta$. **sph↔rect**: $\rho^2=x^2+y^2+z^2$, $\rho\cos\phi=z$, $\rho\sin\phi=r=\sqrt{(x^2+y^2)}$; multiply by ρ to create $\rho\cos\phi$. *Ex* $\rho=2\cos\phi \Rightarrow \rho^2=2\rho\cos\phi \Rightarrow x^2+y^2+z^2=2z$ (sphere, center $(0,0,1)$, radius 1). **Shapes**: $r=c$ cylinder around the z-axis; $\theta=c$ vertical half-plane; $z=c$ horizontal plane; $\rho=c$ sphere; $\phi=c$ cone around the z-axis ($\phi=\pi/4$: $z=\sqrt{(x^2+y^2)}$; $\phi=\pi/2$: xy-plane); $\rho\sin\phi=c \Rightarrow$ cylinder $r=c$; $\rho\cos\phi=c \Rightarrow$ plane $z=c$.

L12 VECTOR-VALUED FUNCTIONS

$\mathbf{r}(t)=f(t)\mathbf{i}+g(t)\mathbf{j}+h(t)\mathbf{k}=\langle f(t),g(t),h(t)\rangle$ **5L1 5L2**: input = one number t (parameter, usually time); output = a vector whose tip traces a curve. f,g,h = **component functions** = the x,y,z coordinates = coefficients of $\mathbf{i},\mathbf{j},\mathbf{k}$. 2 components ⇒ plane curve; 3 ⇒ space curve.

▸**Component functions:** read off the coefficients; constants are fine: $8\mathbf{i}-7\mathbf{j}+(-6t-1)\mathbf{k} \Rightarrow f=8, g=-7, h=-6t-1$. Missing $\mathbf{j} \Rightarrow g=0$. ⚠ Knewton boxes: type only the coefficient ("3", not "3i" — Quiz 3).

▸**Evaluate $\mathbf{r}(a)$:** plug $t=a$ into every component. Any component undefined ($\tan(\pi/2)$, $\sec(\pi/2)$, $\ln 0$, ÷0, $\sqrt{\text{neg}}$) ⇒ $\mathbf{r}(a)$ DNE. *Ex* $\langle 3\tan t, 4\sec t, 5t\rangle$ at $2\pi/3$: $\langle -3\sqrt{3},-8,10\pi/3\rangle$; at $\pi/2$: DNE.

▸**Domain:** intersect every component's domain: $\sqrt{u}$: $u\ge 0$; $1/u$: $u\ne 0$; $\ln u$: $u>0$; tan, sec: $t\ne \pi/2+k\pi$; cot, csc: $t\ne k\pi$ ($k\in\mathbb{Z}$). *Ex* $\langle t^4, \sqrt{(t-8)}, 4/(t-7)\rangle$: $t\ge 8$ (7 already excluded) ⇒ $[8,\infty)$.

▸**Limit 5L3**: limit of each component separately. Plug in first; if 0/0: factor/cancel, conjugate, or **L'Hôpital** ($\lim f/g=\lim f'/g'$). One component DNE ⇒ the whole limit DNE. $t\to\infty$: $e^{-t}\to 0$, $t^n/e^t\to 0$, poly/poly: compare top degrees, $\tan^{-1}t\to\pi/2$, $(\sin t)/t\to 0$. *Ex* $(\sqrt{t}-8)/(t-64)=(\sqrt{t}-8)/[(\sqrt{t}-8)(\sqrt{t}+8)]=1/(\sqrt{t}+8)\to 1/16$ as $t\to 64$. *Ex* $(\sqrt{t}-3)/(t-9)\to 1/6$. $\lim_{u\to 0}(\sin u)/u=1$.

▸**Continuity** at $t=a$: (1) $\mathbf{r}(a)$ exists, (2) $\lim_{t\to a}\mathbf{r}(t)$ exists, (3) they are equal — check every component. "Where is it continuous?" = its domain (polynomials, roots, exp, ln, trig and their combinations are continuous wherever defined). *Ex* $\langle -5\tan t, -3\cos t, -3\sin t\rangle$: $t\ne\pi/2+k\pi$. *Ex* $\langle -2t^2+3t-2, \sqrt{t}/(6t+9), 5t\rangle$: $t\ge 0$.

▸**Graph / recognize:** eliminate t (use $\cos^2+\sin^2=1$). $\langle a\cos t, a\sin t\rangle$ circle radius a (CCW); $\langle a\cos t, b\sin t\rangle$ ellipse; $\langle a\cos t, a\sin t, bt\rangle$ **helix** (circle of radius a in xy, climbing in z); $\langle x_0+at, y_0+bt, z_0+ct\rangle$ line through (x_0,y_0,z_0), direction $\langle a,b,c\rangle$; $\langle t,t^2,t^3\rangle$ twisted cubic. Plot a few t values to get the direction of travel. Segment $P\to Q$: $\mathbf{r}(t)=(1-t)\mathbf{P}+t\mathbf{Q}$, $0\le t\le 1$. ★Curve where 2 surfaces meet: parameterize the simpler one ($x^2+y^2=4 \Rightarrow x=2\cos t, y=2\sin t$), plug into the other for z ($z=xy \Rightarrow z=4\sin t\cos t$).

L13 CALCULUS OF VECTOR FUNCTIONS

▸**Derivative** $\mathbf{r}'(t)=\langle f'(t),g'(t),h'(t)\rangle$ **5L4**: differentiate each component (rules: TOOLKIT). $\mathbf{r}'(t_0)$ = **tangent vector** at the point $\mathbf{r}(t_0)$ (= velocity; points in the direction of motion). Differentiate first, then plug in t. *Ex* $\langle 3\cos t^2, 4\sin t^2, \tan t^2\rangle'=\langle -6t\sin t^2, 8t\cos t^2, 2t\sec^2 t^2\rangle$. *Ex* $\langle e^t\sin t, e^t\cos t, -e^{2t}\rangle'=\langle e^t(\sin t+\cos t), e^t(\cos t-\sin t), -2e^{2t}\rangle$. **Tangent line** at t_0: $\mathbf{L}(s)=\mathbf{r}(t_0)+s\,\mathbf{r}'(t_0)$ **4R13**.

▸**Derivative properties** (c = constant, f = scalar function, $\mathbf{r},\mathbf{u}$ = vector functions): $[c\mathbf{r}]'=c\mathbf{r}'$ **5L5**; $[\mathbf{r}\pm\mathbf{u}]'=\mathbf{r}'\pm\mathbf{u}'$ **5L6**; $[f\mathbf{u}]'=f'\mathbf{u}+f\mathbf{u}'$ **5L7**; $[\mathbf{r}\cdot\mathbf{u}]'=\mathbf{r}'\cdot\mathbf{u}+\mathbf{r}\cdot\mathbf{u}'$ **5L8** (answer is a **scalar** function); $[\mathbf{r}\times\mathbf{u}]'=\mathbf{r}'\times\mathbf{u}+\mathbf{r}\times\mathbf{u}'$ **5R1** (⚠ keep the order; answer is a vector); chain $[\mathbf{r}(f(t))]'=f'(t)\mathbf{r}'(f(t))$ **5R2**: find $\mathbf{r}'$, put $f(t)$ in for t, multiply every component by $f'(t)$. ★For ·, ×, chain: building the combined function first, then differentiating, gives the same answer (often fewer mistakes). *Ex* $\mathbf{r}=\langle -5t^3, e^{8t}, -t^4\rangle$, $\mathbf{u}=\langle -2t, 8\sin t, -3t^2\rangle$: $\mathbf{r}\cdot\mathbf{u}=10t^4+8e^{8t}\sin t+3t^6 \Rightarrow d/dt=40t^3+64e^{8t}\sin t+8e^{8t}\cos t+18t^5$. *Ex* $(d/dt)[-3\mathbf{r}]=-3\mathbf{r}'$. *Ex* $\mathbf{r}=\langle 6e^{3t}, 6e^{-6t}, -4e^{-5t}\rangle$, $f=t^5-3t$: $(5t^4-3)\langle 18e^{3f}, -36e^{-6f}, 20e^{-5f}\rangle$.

▸**Unit tangent** $\mathbf{T}(t)=\mathbf{r}'(t)/\|\mathbf{r}'(t)\|$ **5R3**: unit (length 1) vector in the direction of motion. **How:** $\mathbf{r}'$ ⇒ (plug in t_0 if given) ⇒ $\|\langle a,b,c\rangle\|=\sqrt{(a^2+b^2+c^2)}$ ⇒ divide each component. Needs $\mathbf{r}'\ne\mathbf{0}$. Simplify the norm (factor t^2; $\sin^2+\cos^2=1$). *Ex* $\mathbf{r}=-2t\mathbf{i}+t\mathbf{j}$, $t=3$: $\mathbf{r}'=\langle -2,1\rangle$, $\|\mathbf{r}'\|=\sqrt{5}$, $\mathbf{T}=\langle -2\sqrt{5}/5, \sqrt{5}/5\rangle$. *Ex* $\langle 3t,-5t,-t^3\rangle$ at $t=5$: $\mathbf{r}'=\langle 3,-5,-75\rangle \Rightarrow \mathbf{T}=(1/\sqrt{5659})\langle 3,-5,-75\rangle$.

▸**Integrate 6M1 6M2**: integrate each component. Indefinite: $+\mathbf{C}$, a constant **vector** $\langle C_1,C_2,C_3\rangle$ (write $+\mathbf{C}$ once). Definite: same limits on every component ⇒ a vector of numbers. *Ex* $\int\langle -2\cos 3t, 5\cos 4t\rangle dt=\langle -(2/3)\sin 3t, (5/4)\sin 4t\rangle+\mathbf{C}$. *Ex* $\int_0^{\pi/3}\langle \sin 2t, \tan t, e^{-2t}\rangle dt=\langle 3/4, \ln 2, (1-e^{-2\pi/3})/2\rangle$ (⚠ $\int\tan t\,dt=-\ln|\cos t|$ is not on the sheet). *Ex* $\int_1^3\langle 2t+4, 3t^2-4t, 3t^2-4\rangle dt=\langle 16,10,18\rangle$.

▸**Initial-value problem:** given $\mathbf{r}'$ (or $\mathbf{v}$) and $\mathbf{r}(t_0)$: integrate, plug in t_0, solve each C_i. *Ex* (Quiz 4) $\mathbf{v}=\langle -5t^2-3, -3t^3+1, -2t\rangle$, $\mathbf{r}(0)=\langle -3,0,0\rangle \Rightarrow \mathbf{r}=\langle -(5/3)t^3-3t-3, -(3/4)t^4+t, -t^2\rangle$.

L14 ARC LENGTH & CURVATURE

▸**Arc length** $s=\int_a^b\|\mathbf{r}'(t)\|\,dt$ **6M3** (2D) **6M4** (3D): distance along the curve from $t=a$ to $t=b$; integrand = **speed** $\sqrt{(f'^2+g'^2+h'^2)}$. **How:** $\mathbf{r}'$ ⇒ square and add ⇒ **simplify before taking the root** ($\sin^2+\cos^2=1$; factor out t^2 or e^{2t}; spot perfect squares) ⇒ integrate ⇒ $F(b)-F(a)$. $\sqrt{(t^2)}=|t|=t$ for $t\ge 0$. *Ex* $\langle 3t-2, 4t+5\rangle$, $[0,5]$: speed 5 ⇒ $s=25$. *Ex* $\langle \cos t^2, \sin t^2, t^2\rangle$, $[0,\sqrt{(4\pi)}]$: $\mathbf{r}'=2t\langle -\sin t^2, \cos t^2, 1\rangle$, speed $2\sqrt{2}t \Rightarrow s=\sqrt{2}t^2\Big|_0^{\sqrt{(4\pi)}}=4\sqrt{2}\pi$. *Ex* (Quiz 4) $\langle -t^2, -2t^2\rangle$, $[4,10]$: speed $2\sqrt{5}t \Rightarrow s=\sqrt{5}(100-16)=84\sqrt{5}\approx 187.830$. ★Helix $\langle a\cos t, a\sin t, bt\rangle$: speed $\sqrt{(a^2+b^2)}$.

▸**Arc-length function** $s(t)=\int_a^t\|\mathbf{r}'(u)\|\,du$ **6M5**: distance traveled from the start $t=a$ up to time t (u = dummy variable: write the speed in u, integrate, plug in t and a). $ds/dt=\|\mathbf{r}'(t)\|$ **6L4**. *Ex* $\langle -3\sin t, 7t, -3\cos t\rangle$, $t\ge 0$: speed $\sqrt{(9+49)} \Rightarrow s=\sqrt{58}t$.

▸**Arc-length parameterization $\mathbf{r}(s)$:** 1) speed $\|\mathbf{r}'\|$; 2) $s=\int_{t_0}^t\|\mathbf{r}'(u)\|\,du$ (t_0 = the given start); 3) solve for t in terms of s; 4) substitute into **every** component. Check: $\|d\mathbf{r}/ds\|=1$ and $s=0$ gives the start point. *Ex* helix $\langle \cos t, \sin t, t\rangle$: $s=\sqrt{2}t \Rightarrow \mathbf{r}(s)=\langle \cos(s/\sqrt{2}), \sin(s/\sqrt{2}), s/\sqrt{2}\rangle$. *Ex* $\langle t+3, 2t-4, 2t\rangle$, $t\ge 3$: speed 3, $s=3(t-3)$, $t=3+s/3 \Rightarrow \langle 6+s/3, 2+2s/3, 6+2s/3\rangle$. *Ex* (Quiz 4) $e^{2t}\langle -2,6,-3\rangle$ from $t=0$: speed $14e^{2t}$, $s=7e^{2t}-7$. ★Solve for the block, not t: $e^{2t}=(s+7)/7 \Rightarrow \mathbf{r}(s)=((s+7)/7)\langle -2,6,-3\rangle$. *Ex* $s=5t^2 \Rightarrow t=\sqrt{(s/5)}$. ★Line $\mathbf{r}=\mathbf{P}+t\mathbf{d}$: $\mathbf{r}(s)=\mathbf{r}(t_0)+s\mathbf{d}/\|\mathbf{d}\|$; constant speed c: $t=t_0+s/c$.

▸**Curvature κ:** how sharply the curve bends per unit length ($\kappa\ge 0$; line: 0; circle of radius R: $1/R$; **radius of curvature** $=1/\kappa$). Pick the form that matches what you have: **(a)** $\kappa=\|d\mathbf{T}/ds\|=\|\mathbf{T}'(s)\|$ **5R4**: only when the parameter is arc length s; ★then $\kappa=\|\mathbf{r}''(s)\|$. *Ex* $\langle 15\cos(s/15), 15\sin(s/15)\rangle \Rightarrow \kappa=1/15$. **(b)** $\kappa=\|\mathbf{T}'(t)\|/\|\mathbf{r}'(t)\|$ **5R5**: any t, when you already have $\mathbf{T}$. **(c)** $\kappa=\|\mathbf{r}'\times\mathbf{r}''\|/\|\mathbf{r}'\|^3$ **5R6**: usually easiest for any $\mathbf{r}(t)$ (⚠ cube the speed; ★at a given t plug in first, then cross numeric vectors); 2D: append $z=0 \Rightarrow \kappa=|x'y''-y'x''|/(x'^2+y'^2)^{3/2}$. **(d)** $\kappa=|y''|/[1+(y')^2]^{3/2}$ **5R7**: graph $y=f(x)$; y', y'' = d/dx derivatives evaluated at the x-value. *Ex* $y=3x^2-2x+4$, $x=2$: $y'=10$, $y''=6 \Rightarrow \kappa=6/101^{3/2}\approx 0.00591$. *Ex* (Quiz 4) $y=8\cos 6x$, $x=\pi/6$: $y'=-48\sin\pi=0$, $y''=-288\cos\pi=288 \Rightarrow \kappa=288$. ★Helix $\langle a\cos t, a\sin t, bt\rangle$: $\kappa=a/(a^2+b^2)$ ($a=4, b=3 \Rightarrow 4/25$).

▸**Unit normal N & binormal B:** $\mathbf{N}=\mathbf{T}'(t)/\|\mathbf{T}'(t)\|$ **5R8**: unit vector ⊥ $\mathbf{T}$, pointing to the inside of the bend (toward the osculating circle's center). **How:** get $\mathbf{T}(t)$ simplified as a function of t, pull constant factors out, differentiate, normalize. Needs $\mathbf{T}'\ne\mathbf{0}$ (a line has no $\mathbf{N}$). $\mathbf{B}=\mathbf{T}\times\mathbf{N}$ **6L1**: unit, ⊥ both (cross product **5top**); ★$\mathbf{B}=(\mathbf{r}'\times\mathbf{r}'')/\|\mathbf{r}'\times\mathbf{r}''\|$ (no $\mathbf{N}$ needed). ★$\mathbf{N}$ at one point without $\mathbf{T}'$: $\mathbf{N}=(\mathbf{a}-a_T\mathbf{T})/a_N$ (see L15). ★2D: $\mathbf{N}=\pm\langle -T_2, T_1\rangle$, pick the sign with $\mathbf{N}\cdot\mathbf{r}''>0$. *Ex* helix $\langle \cos t, \sin t, t\rangle$: $\mathbf{T}=(1/\sqrt{2})\langle -\sin t, \cos t, 1\rangle$, $\mathbf{T}'=(1/\sqrt{2})\langle -\cos t, -\sin t, 0\rangle$, $\mathbf{N}=\langle -\cos t, -\sin t, 0\rangle$, $\mathbf{B}=(1/\sqrt{2})\langle \sin t, -\cos t, 1\rangle$. **Planes at** $P=\mathbf{r}(t_0)$ (plane eq. **4R16** with normal $\mathbf{n}$): **osculating** plane contains $\mathbf{T},\mathbf{N}$, $\mathbf{n}=\mathbf{B}$ (or $\mathbf{r}'\times\mathbf{r}''$); **normal** plane contains $\mathbf{N},\mathbf{B}$, $\mathbf{n}=\mathbf{T}$ (or $\mathbf{r}'$). **Osculating circle**: radius $1/\kappa$, center $\mathbf{r}(t_0)+(1/\kappa)\mathbf{N}(t_0)$, lies in the osculating plane. *Ex* helix at $t=\pi/2$: $P=(0,1,\pi/2)$, $\mathbf{B}=(1/\sqrt{2})\langle 1,0,1\rangle \Rightarrow$ osculating plane $x+z=\pi/2$; normal plane ($\mathbf{n}\parallel\mathbf{T}\parallel\langle -1,0,1\rangle$): $-x+z=\pi/2$. $\mathbf{T},\mathbf{N},\mathbf{B}$ = mutually ⊥ unit vectors (TNB frame); $\mathbf{N}=\mathbf{B}\times\mathbf{T}$, $\mathbf{T}=\mathbf{N}\times\mathbf{B}$.

L15 MOTION IN SPACE

▸**Velocity, acceleration, speed:** $\mathbf{v}=\mathbf{r}'$ **6L2** (vector, tangent to the path); $\mathbf{a}=\mathbf{v}'=\mathbf{r}''$ **6L3** (vector); speed $=\|\mathbf{v}\|=\|\mathbf{r}'\|=ds/dt$ **6L4** (scalar ≥ 0). Units: ft, ft/s, ft/s^2. **Distance traveled** $=\int_a^b\|\mathbf{v}\|\,dt$ (arc length, scalar) vs **displacement** $=\mathbf{r}(b)-\mathbf{r}(a)=\int_a^b\mathbf{v}\,dt$ (vector). *Ex* $\langle t^2-3t, 2t-4, t+2\rangle$ at $t=1$: $\mathbf{v}=\langle -1,2,1\rangle$, $\mathbf{a}=\langle 2,0,0\rangle$, speed $\sqrt{6}$. *Ex* $\langle -\sin t, -4\cos t, 0\rangle \Rightarrow \mathbf{v}=\langle -\cos t, 4\sin t, 0\rangle$.

▸**Position from acceleration:** $\mathbf{v}=\int\mathbf{a}\,dt+\mathbf{C}_1$, use $\mathbf{v}(0)$; $\mathbf{r}=\int\mathbf{v}\,dt+\mathbf{C}_2$, use $\mathbf{r}(0)$ (one constant per component). Constant $\mathbf{a}$: $\mathbf{v}=\mathbf{v}_0+\mathbf{a}t$, $\mathbf{r}=\mathbf{r}_0+\mathbf{v}_0t+(1/2)\mathbf{a}t^2$. *Ex* $\mathbf{a}=\langle -7,1,-2\rangle$, $\mathbf{v}(0)=\langle -1,1,1\rangle$, $\mathbf{r}(0)=\mathbf{0} \Rightarrow \mathbf{r}=\langle -(7/2)t^2-t, (1/2)t^2+t, -t^2+t\rangle$.

▸**Tangential & normal components:** $\mathbf{a}=a_T\mathbf{T}+a_N\mathbf{N}$ **6R2** $=v'\mathbf{T}+v^2\kappa\mathbf{N}$ **6L5** (v = speed, v' = its rate of change); $\mathbf{a}$ always lies in the osculating plane. a_T = how fast the speed changes (+ speeding up, − slowing, 0 = constant speed): $a_T=\mathbf{a}\cdot\mathbf{T}=(\mathbf{v}\cdot\mathbf{a})/\|\mathbf{v}\|$ **6L6**. a_N = turning (centripetal, ≥ 0): $a_N=\mathbf{a}\cdot\mathbf{N}=\|\mathbf{v}\times\mathbf{a}\|/\|\mathbf{v}\|=\sqrt{(\|\mathbf{a}\|^2-a_T^2)}$ **6R1** $=\kappa v^2$. **How:** $\mathbf{v},\mathbf{a}$ (plug in t if given) ⇒ $\mathbf{v}\cdot\mathbf{a}$ and $\|\mathbf{v}\|$ ⇒ a_T ⇒ ★a_N by the root form (no cross product). $\|\mathbf{a}\|^2=a_T^2+a_N^2$. ★Constant speed ⟺ $\mathbf{v}\cdot\mathbf{a}=0$ ($\mathbf{v}\perp\mathbf{a}$). *Ex* $\langle 8t^3-5, -2t^2, -8t-3\rangle$: $\mathbf{v}=\langle 24t^2, -4t, -8\rangle$, $\mathbf{a}=\langle 48t, -4, 0\rangle \Rightarrow a_T=(1152t^3+16t)/\sqrt{(576t^4+16t^2+64)}$. *Ex* (Quiz 4) $\langle 5t-9, 3-6t, -2t\rangle$: $\mathbf{a}=\mathbf{0} \Rightarrow a_T=0$. *Ex* $\langle -4\cos(4\pi t/3), 4\sin(4\pi t/3)\rangle$: speed $16\pi/3$ constant ⇒ $a_T=0$, $a_N=v^2/R=64\pi^2/9$ (any t).

▸**Friction / centripetal force:** $F=m\,a_N=mv^2/R$ (m = mass in kg, v = speed in m/s, R = turn radius in m, $\kappa=1/R$; F in N). *Ex* 2197 kg, 28 m/s, $R=19$ m ⇒ $F=(2197\cdot 784)/19\approx 90{,}655$ N.

▸**Projectile 6B**: launch angle θ above horizontal, initial speed v_0, launch height h (⚠ the sheet's "x_0" in the $\mathbf{j}$ part is h), $g=9.8$ m/s^2 or 32 ft/s^2: $x(t)=(v_0\cos\theta)t$, $y(t)=h+(v_0\sin\theta)t-(1/2)gt^2$, $v_y=v_0\sin\theta-gt$. **Max height**: $v_y=0 \Rightarrow t_{top}=(v_0\sin\theta)/g$, $y_{max}=h+(v_0\sin\theta)^2/(2g)$. **Lands** when $y=0$ (quadratic formula, + root): $t=(v_0\sin\theta+\sqrt{((v_0\sin\theta)^2+2gh)})/g$. **Range** $=x(t_{land})$. $h=0$: $t=(2v_0\sin\theta)/g$, range $(v_0^2\sin 2\theta)/g$; solve backwards for v_0 or θ from a given range. *Ex* 600 ft/s, 30°, 100-ft cliff: $v_0\sin\theta=300$, $y_{max}=100+300^2/64=1506.25$ ft; $16t^2-300t-100=0 \Rightarrow t\approx 19.08$ s; range $\approx 300\sqrt{3}(19.08)\approx 9913$ ft. *Ex* 156 m, 265 m/s, 49° ⇒ $t\approx 41.6$ s, range ≈ 7229 m.

L16 FUNCTIONS OF SEVERAL VARIABLES

$z=f(x,y)$: input = a point (x,y) in the domain $D\subseteq\mathbb{R}^2$; output = one number z (a height); graph = a surface. $w=f(x,y,z)$: 3 inputs (domain = region of space).

▸**Domain:** every input where the formula works; intersect all restrictions: denominator ≠ 0; even-root radicand ≥ 0; root in a denominator > 0; $\ln u$: $u>0$; $\sin^{-1}u, \cos^{-1}u$: $-1\le u\le 1$. Polynomials, sin, cos, e^u: all of $\mathbb{R}^2$. Answer as an inequality / $\{(x,y)\mid\ldots\}$ (Knewton: type it; $\mathbb{R}^2$ if everything). Sketch: boundary = the equality curve (solid if included, dashed if not); test a point. *Ex* (Quiz 4) $\sqrt{(3x^2+5y^2-15)}$: $3x^2+5y^2\ge 15$ (outside an ellipse). *Ex* $\sqrt{(9x^2-5y^2-45)}$: $9x^2-5y^2\ge 45$. *Ex* $xy/(-7x^2+8y)$: $y\ne(7/8)x^2$. *Ex* $\cos\sqrt{(-6x-9y)}$: $2x+3y\le 0$. *Ex* $(x-y-z)/\sqrt{(6-x^2-3y^2-9z^2)}$: $x^2+3y^2+9z^2<6$.

▸**Range:** all possible outputs: bound each piece (squares ≥ 0; $\sqrt{\ }\ge 0$; $\sqrt{(c-\ldots)}\le\sqrt{c}$) and check the ends are actually reached. *Ex* $-4x^2-9y^2$: $(-\infty,0]$. *Ex* $-1-\sqrt{(36-y^2)}$: root $\in[0,6] \Rightarrow [-7,-1]$. *Ex* $6-\sqrt{(x^2+y^2-25)}$: $(-\infty,6]$. [ ] includes the endpoint, ( ) excludes; ∞ always gets (.

▸**Evaluate:** plug in carefully: $-5x^2-2y-4$ at $(-5,1)$: $-5(25)-2-4=-131$ (square first, then multiply by −5).

▸**Level curve** for value c: set $f(x,y)=c$, simplify ⇒ a curve in the xy-plane (contour). Contour map = many level curves; close together = steep. Families: $x^2+y^2=c$ circles; $x^2-y^2=c$ or $xy=c$ hyperbolas; $ax+by=c$ parallel lines; $y-x^2=c$ parabolas. *Ex* $3x+7y-7=2 \Rightarrow 3x+7y=9$. *Ex* (Quiz 4) $8x^2+5y^2=2$ (ellipse). *Ex* $-2+\sin(-7x-7y)=-1 \Rightarrow \sin(-7x-7y)=1 \Rightarrow -7x-7y=\pi/2+2k\pi \Rightarrow y=-x-\pi/14+2k\pi/7$, $k\in\mathbb{Z}$. General solutions: $\sin u=1$: $u=\pi/2+2k\pi$; $\sin u=-1$: $3\pi/2+2k\pi$; $\sin u=0$: $k\pi$; $\cos u=1$: $2k\pi$; $\cos u=0$: $\pi/2+k\pi$; $\cos u=-1$: $\pi+2k\pi$.

▸**Vertical trace:** fix an **input**: $x=a \Rightarrow z=f(a,y)$ (curve in the plane $x=a$, ∥ yz-plane); $y=b \Rightarrow z=f(x,b)$ (∥ xz-plane). A level curve fixes the **output** $z=c$. *Ex* $6x+8y^2$, $x=5 \Rightarrow z=30+8y^2$. *Ex* $-5\cos 4x\sin 3y$, $y=\pi/6 \Rightarrow z=-5\cos 4x$. *Ex* (Quiz 4) $7x^2-5y$, $y=0 \Rightarrow z=7x^2$.

▸**Level surfaces & graphs:** $f(x,y,z)=c$ is a surface (e.g. $x^2+y^2+z^2=c$: spheres). $z=\sqrt{(R^2-x^2-y^2)}$ upper hemisphere; $z=\sqrt{(x^2+y^2)}$ cone; $z=x^2+y^2$ paraboloid; $z=ax+by+c$ plane (plot the intercepts).

L17 PARTIAL DERIVATIVES

▸**First partials:** $f_x=\partial f/\partial x$: differentiate in x treating y (and z) as **constants**; f_y: treat x (and z) as constants. $f_x=\lim_{h\to 0}(f(x+h,y)-f(x,y))/h$ = slope of the surface in the x-direction. Frozen variables act like numbers: $\partial_x(3xy)=3y$; $\partial_x(2y^2)=0$; $\partial_x e^{xy}=ye^{xy}$; $\partial_y(x^2y^3)=3x^2y^2$. Chain rule: $\partial_x\sin(x^2y-2x+4)=(2xy-2)\cos(\cdot)$, $\partial_y=x^2\cos(\cdot)$; (Quiz 4) $\partial_y\tan(x+7y^3)=21y^2\sec^2(x+7y^3)$. Product rule only when **both** factors contain the variable: $\partial_x[e^{2xy}\sin 2x\sin 2y]=2ye^{2xy}\sin 2x\sin 2y+2e^{2xy}\cos 2x\sin 2y$. Quotient: $\partial_y(u/v)=(u_yv-uv_y)/v^2$. ★Implicit ($z$ defined by $F(x,y,z)=0$): $z_x=-F_x/F_z$, $z_y=-F_y/F_z$ (p6 Unit 3 block), or differentiate both sides in x treating z as $z(x,y)$ and solve for z_x. ★"Show f satisfies $f_{xx}+f_{yy}=0$": compute those partials, substitute, simplify.

▸**3+ variables:** f_x, f_y, f_z: freeze the other two. *Ex* $6y\sin(9x^4z^6+2y^3)$: $f_x=216x^3yz^6\cos(\cdot)$; $f_y=6\sin(\cdot)+36y^3\cos(\cdot)$ (product!); $f_z=324x^4yz^5\cos(\cdot)$.

▸**Higher order:** $f_{xx}=(f_x)_x$, $f_{yy}=(f_y)_y$, $f_{xy}=(f_x)_y$ (⚠ subscripts read left⇒right: x first), $f_{yx}=(f_y)_x$; operator form $\partial^2 f/\partial y\partial x=f_{xy}$ (rightmost first). From a table/contour map: $f_x(a,b)\approx(f(a+h,b)-f(a,b))/h$ (average the forward and backward estimates). **Clairaut**: mixed partials continuous ⇒ $f_{xy}=f_{yx}$ (★do the easier order; use the other as a check). Third order: $f_{yyx}=((f_y)_y)_x$. *Ex* (Quiz 4) $3x+5y^7+6xy-2y^9$: $f_y=35y^6+6x-18y^8$, $f_{yy}=210y^5-144y^7$. *Ex* $5x^2-4xy^8+3xy-1$: $f_{yyx}=-224y^6$. *Ex* $(3x^5-3y^5-3)^4$: $f_x=60x^4(\cdot)^3$, $f_{xy}=-2700x^4y^4(\cdot)^2$. *Ex* $3e^{8x}\cos(2yz^2)$: $f_{xyz}=-96ze^{8x}\sin(2yz^2)-192yz^3e^{8x}\cos(2yz^2)$.

L18 TANGENT PLANES, LINEARIZATION, DIFFERENTIALS

▸**Tangent plane** to $z=f(x,y)$ at (x_0,y_0): $z=f(x_0,y_0)+f_x(x_0,y_0)(x-x_0)+f_y(x_0,y_0)(y-y_0)$ **6R3**. **How:** 1) $z_0=f(x_0,y_0)$; 2) f_x, f_y as formulas; 3) plug in (x_0,y_0) ⇒ two numbers (slopes in the x- and y-directions); 4) assemble (simplify to $z=Ax+By+C$ only if asked). Horizontal tangent plane ⟺ $f_x=f_y=0$. Normal vector $\langle f_x, f_y, -1\rangle$; normal line $\langle x_0,y_0,z_0\rangle+t\langle f_x, f_y, -1\rangle$. ★Surface $F(x,y,z)=k$: $F_x(x-x_0)+F_y(y-y_0)+F_z(z-z_0)=0$. *Ex* $2x^2-3xy+8y^2+2x-4y+4$ at $(2,-1)$: $f=34$, $f_x=4x-3y+2=13$, $f_y=-3x+16y-4=-26 \Rightarrow z=34+13(x-2)-26(y+1)$. *Ex* $\sin 2x\cos 3y$ at $(\pi/3,\pi/4)$: $f=-\sqrt{6}/4$, $f_x=2\cos 2x\cos 3y=\sqrt{2}/2$, $f_y=-3\sin 2x\sin 3y=-3\sqrt{6}/4$. *Ex* $-3xy/(10x+3y)$ at $(-2,-2)$: $z=6/13-(9/169)(x+2)-(30/169)(y+2)$.

▸**Linear approximation** $L(x,y)$ **6R4** = the same formula; $f(x,y)\approx L(x,y)$ near (x_0,y_0). **How:** base point = a nearby "nice" point where f is easy; plug the target point into L. 3 variables: add $f_z(x_0,y_0,z_0)(z-z_0)$ (same for dw). *Ex* $f=-5x^3-2y^2-3xy$, base $(1,2)$, find $f(0.996,1.9)$: $f=-19$, $f_x=-15x^2-3y=-21$, $f_y=-4y-3x=-11 \Rightarrow L=-19-21(-0.004)-11(-0.1)=-17.82$. *Ex* $9\cos x+10\sin(x+y)$ at $(0,0)$: $L=9+10x+10y \Rightarrow L(0.003,-0.003)=9$.

▸**Differentiability:** f is differentiable at P when the tangent plane fits: $f=L+E$ with $E/\sqrt{((x-x_0)^2+(y-y_0)^2)}\to 0$. **Test**: f_x, f_y exist near P and are continuous at P ⇒ differentiable. One-way chain: continuous partials ⇒ differentiable ⇒ continuous (⚠ converses are false). ★Sums, products and compositions of polynomials, exp and trig are differentiable wherever defined, e.g. $e^{5x-4y}\cos(-3x-5y)$: partials continuous everywhere ⇒ differentiable at $(0,0)$.

▸**Total differential** $dz=f_x(x_0,y_0)\,dx+f_y(x_0,y_0)\,dy$ **6R5**: $dx=\Delta x$, $dy=\Delta y$ = small (signed) input changes; $dz\approx\Delta z=f(x_0+\Delta x, y_0+\Delta y)-f(x_0,y_0)$ (the true change). "Find dz" ⇒ leave f_x, f_y as formulas; "approximate Δz" ⇒ plug in numbers. New value $\approx f(x_0,y_0)+dz$. Error propagation: max error $\approx|f_x||\Delta x|+|f_y||\Delta y|$. *Ex* $z=-3xy/(x-y)$: $dz=(3y^2/(x-y)^2)dx-(3x^2/(x-y)^2)dy$. *Ex* $-3x^2+4y^2+5xy$ at $(2,-1)$, $\Delta x=0.02$, $\Delta y=-0.01$: $f_x=-6x+5y=-17$, $f_y=8y+5x=2 \Rightarrow dz=-0.34-0.02=-0.36$. *Ex* can $V=\pi r^2h$, $r=3$, $h=10$, errors 0.05, 0.1: $dV=2\pi rh\,dr+\pi r^2dh=3\pi+0.9\pi\approx 12.25$. Relative error dV/V (×100 for %).

BY-HAND TOOLKIT (no CAS)

▸**Derivatives.** On **p3** (left): k, x^n, $\ln|x|$, e^x, e^{bx}, sin, cos (also of bx), tan, csc, sec, cot, $\sin^{-1}$, $\tan^{-1}$, $\sec^{-1}$. **Product** $(fg)'=f'g+fg'$. **Quotient** $(f/g)'=(f'g-fg')/g^2$. **Chain** $[f(g)]'=f'(g)\cdot g'$ (outside derivative with the inside untouched, times inside derivative). Chain forms (u = inside): $(u^n)'=nu^{n-1}u'$; $(\sqrt{u})'=u'/(2\sqrt{u})$; $(1/u)'=-u'/u^2$; $(e^u)'=e^uu'$; $(a^u)'=a^u\ln a\,u'$; $(\ln u)'=u'/u$; $(\log_a u)'=u'/(u\ln a)$; $(\sin u)'=u'\cos u$; $(\cos u)'=-u'\sin u$; $(\tan u)'=u'\sec^2u$; $(\sec u)'=u'\sec u\tan u$; $(\cos^{-1}u)'=-u'/\sqrt{(1-u^2)}$; $(\cot^{-1}u)'=-u'/(1+u^2)$. Rewrite roots and fractions as powers first ($4/t^2=4t^{-2}$).

▸**Integrals.** On **p3** (right): k, x^n, $1/x$, e^x, e^{bx}, $\cos bx$, $\sin bx$, $\sec^2$, csc cot, sec tan, $\csc^2$, $1/\sqrt{(1-x^2)}$, $1/(1+x^2)$. Not on the sheet: $\int\tan u=-\ln|\cos u|=\ln|\sec u|$; $\int\cot u=\ln|\sin u|$; $\int\sec u=\ln|\sec u+\tan u|$; $\int\csc u=-\ln|\csc u+\cot u|$; $\int\sec^2bx=(1/b)\tan bx$; $\int a^x=a^x/(\ln a)$; $\int\ln x=x\ln x-x$; $\int(ax+b)^n=(ax+b)^{n+1}/(a(n+1))$; $\int dx/(ax+b)=(1/a)\ln|ax+b|$; $\int dx/(a^2+x^2)=(1/a)\tan^{-1}(x/a)$; $\int\sin^2x=x/2-(\sin 2x)/4$, $\int\cos^2x=x/2+(\sin 2x)/4$ (power-reducing **p1**). **u-sub**: u = inside, $du=u'\,dt$, replace everything; definite: change the limits too. **Parts**: $\int u\,dv=uv-\int v\,du$ (u by LIATE: Log, Inverse trig, Algebraic, Trig, Exp). $\int_a^b f=F(b)-F(a)$.

▸**Vectors.** Magnitude $\|\langle a,b,c\rangle\|=\sqrt{(a^2+b^2+c^2)}$ **4L11**; unit vector $\mathbf{v}/\|\mathbf{v}\|$ **4L12**. **Dot** $\mathbf{u}\cdot\mathbf{v}=u_1v_1+u_2v_2+u_3v_3$ **4L16** (a scalar; $=0$ ⟺ ⊥); angle $\cos\theta=(\mathbf{u}\cdot\mathbf{v})/(\|\mathbf{u}\|\|\mathbf{v}\|)$ **4R1**. **Cross 5top**: $\mathbf{u}\times\mathbf{v}=\langle u_2v_3-u_3v_2, u_3v_1-u_1v_3, u_1v_2-u_2v_1\rangle$. **How:** rows $\mathbf{i}\,\mathbf{j}\,\mathbf{k}$ / $\mathbf{u}$ / $\mathbf{v}$; for each unit vector cover its column and take the 2 × 2 determinant $ad-bc$; ⚠ the middle ($\mathbf{j}$) term gets a minus. Result ⊥ both (check: dot with each = 0); $\mathbf{v}\times\mathbf{u}=-(\mathbf{u}\times\mathbf{v})$; $\|\mathbf{u}\times\mathbf{v}\|$ = parallelogram area; 2D vectors: append 0. ★Factor scalars out first: $(c\mathbf{u})\times(d\mathbf{v})=cd(\mathbf{u}\times\mathbf{v})$, $\|c\mathbf{v}\|=|c|\|\mathbf{v}\|$.

▸**Algebra & trig.** Complete the square: $ax^2+bx=a(x+b/(2a))^2-b^2/(4a)$. Quadratic formula $x=(-b\pm\sqrt{(b^2-4ac)})/(2a)$. Rationalize $a/\sqrt{b}=a\sqrt{b}/b$. $\sqrt{(a^2b)}=|a|\sqrt{b}$; $\sqrt{(x^2)}=|x|$. $\sin^2+\cos^2=1$, $1+\tan^2=\sec^2$ **p1**. Logs: $e^{\ln a}=a$, $\ln e^k=k$, $\ln ab=\ln a+\ln b$; $e^{2t}=c \Rightarrow t=(1/2)\ln c$. Degrees = radians×180/π. Unit circle **p2**. ★Perfect-square speeds: $4t^2+4+1/t^2=(2t+1/t)^2$; $e^{2t}+2+e^{-2t}=(e^t+e^{-t})^2$; $\langle e^t\cos t, e^t\sin t, e^t\rangle$: speed $\sqrt{3}e^t$.

▸**Answer checks.** Vector or scalar? Exact or decimal (round only at the end)? Degrees vs radians (calculator mode)? $+\mathbf{C}$ written once? Given θ kept? Only coefficients in $\mathbf{i}/\mathbf{j}/\mathbf{k}$ boxes? κ, a_N, speed, ρ, r all ≥ 0?

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